Kutty Banerjee

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SUMMARY

Hands on Technical Leader with over 15 years experience working on different parts of the GPU stack, Kernel-Firmware Drivers, Low Latency Runtimes and End to End Performance Analysis. 18 US Patents in flight in the above areas. Mentored and promoted engineers, built, lead and grew teams, delivered large scale projects on time.

EXPERIENCE

Engineering Manager, Google

Moutain View, CA — Dec 2019 - Present

Managing the team responsible for scheduling Machine Learning Inference and Training workloads on Google's Data Center across thousands of machines using TPUs. The primary objectives are to minimize inference latency, scale out training and maximize Data Center TPU Utilization.

Engineering Manager, Apple

Cupertino, CA — Aug 2013 - Dec 2019

- Part of the founding team involved in the design and implementation of Apple's homegrown GPU.
- **Managed** the **GPU Driver Scheduling** team, a vertically integrated team involving UMD/KMD and Firmware.
- Team was responsible for policies involving Command scheduling, Resource Management, Multitenancy, Quality of Service and End-to-End Performance.
- Team scaled the iOS mobile GPU Driver to work optimally on ARM based macOS desktops at the same time delivering **low latency** AR/VR solutions on mobile SoCs.
- **Built the team and lead the following projects** while influencing future generation GPU/SoC hardware

Driver Scaling

- Scaled the iOS GPU KMD Driver and Firmware to sustain 1000 render passes from 5 applications @120Hz for an undisclosed product.
- This required rewriting the **Driver locking strategy**, changes to XNU Kernel Thread scheduling, **Partitioning of Render Graphs** and rewriting the **Firmware scheduler**. Multiple patents in flight in this area.

Low Latency Graphics

- Managed the team and lead a Cross-Functional effort called **GPU Responsiveness** to reduce End-to-End Driver latency for AR solutions.
- Notable contributions include reducing Firmware command selection overhead from **21us** to **5us** on the same generation of hardware and influencing future generation hardware to do **sub-100ns** selection.
- Converted Graphics Render/Compute passes into a Graph for low latency Firmware processing.
- Performed in-memory Graph rewrites to do optimal Command scheduling on the GPU to maximize GPU utilization and minimize Intra-GPU Network traffic.

Multitenancy and Quality of Service

- Designed and implemented Quality of Service in Metal
- Designed hardware changes to provide End-to-End quality of service inside the GPU Core, cache, memory bandwidth, utilization and improved context switch latency.
- This allows running multiple Applications on the GPU thus increasing its utilization while guaranteeing Quality of Service.
- Team provided both **Deadline** and **Utilization based schedulers** based on the target platform.
- Delivered Visualization Tools to detect QoS problems quickly.

End to End Performance

- Lead the effort to implement a cycle accurate **Driver Performance Simulator** to analyze the impact of latency at various levels

of the Software stack.
- This allowed projecting and improving performance bottlenecks on future hardware with future application patterns and assessing the **Value Proposition for multi-quarter multi-engineer projects**.

Software Architect, NVIDIA

Santa Clara, CA — May 2005-May 2013

Lead the Resource Manager Driver development for Fermi. Was part of NVIDIA's first native ARMv8 64-bit CPU Implementation called Denver.

ARMv8 Denver CPU Runtime Compiler
- Was part of the Just-in-Time compiler runtime team that did Dynamic Binary Translation of ARMv8 to a native VLIW ISA to extract Instruction Level Parallelism and Memory Level Parallelism.
- Runtime profiled and replaced branches with trap-on-branch instructions to help compact trace scheduling
- Added rollback for this optimization if traps were taken frequently in a small windows of time.
- Implemented Branch Reprofiling to detect dynamic changes in branch weights of a running program and trigger recompilation. Improved libquantum SPECFP by 45%
- Did fullstack Linux optimizations for Denver. Implemented ARMv7 Large Page Support FreeBSD Super Pages style.
- Added a new Thread Local Memory type with support in madvise and mmap system calls. Memory Descriptors marked Thread Local are treated separately in the Kernel MM. TLB invalidates, IPI broadcasts and cache snooping for these ranges can be elided on processors where doing so helps performance. Useful for Javascript Thread Local Heaps.

Resource Manager : Maxwell UVM Architect
- Was one of the founding members working on the CUDA UVM project. Did scoping of Linux kernel changes required to implement Shadow Page Tables. Worked with HW teams to explore Page Fault and Fault Aggregation support on GPU.

Resource Manager : Fermi Chip Lead
- As the Chip Lead for the Fermi GPU Generation, lead a team of 8 engineers and oversaw development across Simulation, RTL, Emulation, Bringup and Product Release.
- Role included Task Assignment, Initial Triage, Interaction with Hardware and other Software teams, mentoring new Engineers and Code Review. Responsible for overall progress of the Kernel team for Fermi.
- Was the first engineer to start work on the Kernel/Firmware for Fermi. Wrote and maintained large parts of the RM code including Thread Scheduling, Interrupt Handling, Memory Management, Page Table Setup, Context Switch Support using Firmware, Cache mgmt., TLB invalidate batching.

Resource Manager : Fermi Command Buffer Scheduling Lead
- Implemented Command Buffer Scheduler in the Driver. Fermi added Hardware support for DX11 Per Node Command Buffers. Replaced the old Time Slice Based Scheduler with Per Node Queues.
- Implemented a Priority based Timeslice mechanism.
- Implemented a Non Stalling Lock Free Interrupt Tree for routing DXG kernel notifiers without delay

EDUCATION

MS Computer Science, Worcester Polytechnic Institute

Worcester, MA — Aug 2003 - May 2005

- Thesis: Remote Execution of 3D Graphics for Mobile Systems.
- Recipient of WPI Institute Fellowship Award for outstanding research work.
- Several Journal and Conference Publications

BE Computer Science, University of Mumbai

Mumbai, India — July 1999 - May 2003